

Notes: Greenwood & Vayanos (RFS, 2014)

Bond Supply and Excess Bond Returns

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Main sample and prewar sample	4
3.2	Cash-flow decomposition	4
3.3	Main supply measures	4
3.4	Outcome variables	5
3.5	Macroeconomic and risk controls	5
4	Model and empirical specifications	5
4.1	Economic mechanism	5
4.2	Yield regression	6
4.3	Return regression	6
4.4	Maturity-gradient hypothesis	6
4.5	Instrumental-variables specification	6
4.6	Arbitrageur-wealth interaction	7
5	Identification and empirical design	7
5.1	What the paper is trying to identify	7
5.2	Predetermined supply and face-value measurement	7
5.3	Short-rate control	8
5.4	Instrumental variables design	8
5.5	Historical validation using the prewar sample	8
5.6	Maturity-gradient validation	8
5.7	Arbitrageur capital validation	9
5.8	Threats to identification	9
6	Tables and figures	9
6.1	Figure 1: Principal and coupon payments; Figure 2: Bond supply, 1952–2007	9
6.2	Table 1: Summary Statistics; Figure 3: Bond supply, 1916–1940	10
6.3	Table 2: Bond supply, bond yields, and bond returns; Figure 4: Bond supply and excess bond returns	11
6.4	Table 3: Regression coefficients as a function of maturity; Figure 5: Regression coefficients as a function of maturity	12
6.5	Table 4: Instrumental-variables tests; Table 5: Robustness tests	14
6.6	Table 6: Bond supply, yields, and returns, 1916–1940; Table 7: Arbitrageur wealth	15

6.7	Figure 6: Model calibration	15
7	Short summary	16
8	Conclusion	16
8.1	Core conclusion	16
8.2	Economic intuition	17
8.3	Identification takeaway	17
8.4	Policy implication	17
8.5	Limitations	17
8.6	Takeaway	18

1 Big picture

- **Question.** Do the supply and maturity structure of U.S. government debt affect bond yields and expected bond returns?
- **Why this matters.** In representative-agent frictionless models, Ricardian equivalence implies that the maturity structure of government debt should not matter for prices. Portfolio-balance and preferred-habitat views instead imply that when investors must absorb more duration risk, long yields and expected returns should rise.
- **Core mechanism.** Risk-averse arbitrageurs absorb shocks to demand and supply across maturities. A larger supply of duration forces them to bear more interest-rate risk, so they require higher expected excess returns. Prices fall and yields rise.
- **Empirical object.** The key supply variable is *maturity-weighted debt to GDP*, denoted MWD/GDP . It weights each promised Treasury cash flow by its remaining maturity and scales by GDP.
- **Main result.** Higher MWD/GDP predicts higher bond yields and higher future bond returns, controlling for the one-year yield.
- **Cross-maturity result.** Effects are stronger at longer maturities, especially for future returns. This matches the model because longer bonds have larger exposure to duration risk.
- **Time-series result.** The supply-return relation is stronger after arbitrageurs have recently lost money, consistent with risk-bearing capacity or effective risk aversion varying over time.
- **Identification strategy.** The paper combines time-series predictive regressions, long historical samples, IV regressions that instrument maturity-weighted debt with debt-to-GDP, robustness controls, and model calibration.
- **Economic takeaway.** Government debt supply is not only a fiscal accounting object. Its maturity structure changes the amount of duration risk held by private investors, and this risk-bearing channel affects bond risk premia.

2 Incremental contribution of the paper

- **From event studies to long-run time series.** Earlier work often studies particular operations, such as Operation Twist, Treasury buybacks, or QE announcements. This paper studies lower-frequency variation in the aggregate maturity structure of Treasury debt over a long time series.
- **A theoretically motivated supply measure.** Instead of using only debt-to-GDP or long-term debt shares, the paper constructs a duration-weighted measure, MWD/GDP , that is closely aligned with the duration risk arbitrageurs must absorb.
- **Integrated theory and empirics.** The empirical tests are organized around a term-structure model that yields clear predictions for yields, expected returns, maturity patterns, and time variation with arbitrageur risk capacity.
- **Separation of yields and expected returns.** The paper emphasizes that the effect of supply on current yields is smaller than the effect on current expected returns because yields average expected returns over a bond's remaining life.
- **Historical validation.** The paper constructs a second prewar sample, 1916–1940, and finds broadly similar patterns, which reduces the risk that the result is special to the postwar period.
- **Endogeneity response.** The paper uses marketable debt-to-GDP as an instrument for maturity-weighted debt, arguing that debt-to-GDP is driven largely by accumulated deficits rather than by current investor demand for maturity.

- **Connection to central-bank asset purchases.** The results help interpret QE and maturity-extension policies: removing long-duration bonds from the market should lower duration risk borne by arbitrageurs and therefore lower long-term yields and risk premia.

3 Data and measurement

3.1 Main sample and prewar sample

- The main sample runs from June 1952 to December 2007. It ends in 2007 because the paper studies future three-year returns that extend through 2010.
- The paper also builds a prewar sample from June 1916 to June 1940.
- The period between 1941 and 1951 is omitted because Federal Reserve policies pegged yields across the term structure, leaving little meaningful yield variation.
- Main Treasury security data come from the CRSP historical government bond database. The paper uses bond characteristics, coupon rates, maturities, and face values outstanding.

Interpretation: The two-sample design is important. The postwar sample provides rich monthly CRSP security-level information, while the prewar sample tests whether the central finding is historically robust.

3.2 Cash-flow decomposition

For each government bond, the paper breaks total promised cash flows into coupon payments and principal repayments. For each future remaining maturity τ , the paper constructs:

$$D_t^{(\tau)} = \text{face-value cash flow due at maturity } \tau.$$

- Coupon payments are assigned to their payment dates.
- Principal is assigned to the maturity date.
- The paper groups maturities beyond 30 years into a 30-year bucket.
- Supply is measured in face-value rather than market-value terms to avoid a mechanical feedback from yields into the market value of bonds.

Interpretation: The key empirical variable is not the stock of Treasury debt alone; it is the whole future cash-flow profile of that debt. The model says arbitrageurs care about the duration risk embedded in that profile.

3.3 Main supply measures

The primary supply measure is maturity-weighted debt to GDP:

$$\left(\frac{MWD}{GDP}\right)_t = \frac{\sum_{0 < \tau \leq 30} D_t^{(\tau)} \tau}{GDP_t}.$$

A second supply measure is long-term debt to GDP:

$$\left(\frac{LTD}{GDP}\right)_t = \frac{\sum_{10 \leq \tau \leq 30} D_t^{(\tau)}}{GDP_t}.$$

The paper also reports:

$$M_t = \text{dollar-weighted average maturity}, \quad \frac{D_t}{GDP_t} = \text{aggregate principal debt to GDP}.$$

Interpretation: MWD/GDP combines the level of debt and its maturity structure. A government can raise MWD/GDP by issuing more debt, issuing longer-maturity debt, or both.

3.4 Outcome variables

- $y_t^{(\tau)}$: yield on a bond with maturity τ .
- $y_t^{(1)}$: one-year yield, used as a control for the short rate.
- $r_{t+1}^{(\tau)}$: one-year return on a bond with maturity τ .
- $r_{t+k}^{(\tau)}$: cumulative k -year return on a bond whose maturity rolls down over time.
- Long-term bond returns are used for the main one-, three-, and five-year return tests.

Interpretation: The distinction between yields and future returns is central. Yields measure prices today. Future returns test whether supply forecasts bond risk premia.

3.5 Macroeconomic and risk controls

The robustness tests include controls for:

- output gap;
- output growth;
- inflation;
- inflation risk;
- interest-rate risk;
- stock-market risk;
- term spread;
- time trend;
- future changes in debt maturity;
- Federal Reserve holdings.

Interpretation: These controls address the concern that supply might proxy for business-cycle risk, inflation risk, term-spread information, or central-bank balance-sheet operations rather than a distinct duration-supply channel.

4 Model and empirical specifications

4.1 Economic mechanism

The model can be summarized as:

Higher duration-weighted Treasury supply $\Rightarrow$ more duration risk held by arbitrageurs $\Rightarrow$ higher price of duration risk $\Rightarrow$ higher

Interpretation: The model is a portfolio-balance theory with a modern asset-pricing foundation. Supply matters only because arbitrageurs are risk averse and cannot costlessly absorb duration risk.

4.2 Yield regression

The baseline yield regression is:

$$y_t^{(\tau)} = a + bX_t + cy_t^{(1)} + u_t,$$

where X_t is either MWD/GDP or LTD/GDP .

- A positive b means that higher duration supply is associated with higher yields.
- The one-year yield controls for the short rate.
- The model predicts that b should be positive and either increasing or hump-shaped in maturity.

Interpretation: The yield regression asks whether government debt maturity affects the current term structure beyond what is captured by the short rate.

4.3 Return regression

The baseline return regression is:

$$r_{t+k,k}^{(\tau)} = a + bX_t + cy_t^{(1)} + u_{t+k}.$$

- A positive b means that higher supply predicts higher future bond returns.
- The paper studies one-, three-, and five-year returns.
- The model predicts that supply effects on expected returns should be stronger for longer maturities.

Interpretation: The return regression is the stronger risk-premium test. If higher supply predicts high future returns, then supply is not merely shifting yields through expectations of short rates; it is associated with compensation for bearing risk.

4.4 Maturity-gradient hypothesis

The model implies:

$$\frac{\partial E_t[r_{t+1}^{(\tau)}]}{\partial X_t} \text{ increases with } \tau.$$

For yields, the effect is smaller:

$$\left| \frac{\partial y_t^{(\tau)}}{\partial X_t} \right| < \left| \frac{\partial E_t[r_{t+1}^{(\tau)}]}{\partial X_t} \right|.$$

Interpretation: Long bonds have more duration exposure, so a shock to the price of duration risk affects them more. Yields are averages of expected short-rate and risk-premium components over the bond's life, so yield effects are muted relative to immediate expected-return effects.

4.5 Instrumental-variables specification

The IV first stage instruments maturity-weighted debt with debt-to-GDP:

$$X_t = \alpha + \phi \left(\frac{D}{GDP} \right)_t + \delta y_t^{(1)} + u_t.$$

The second-stage regressions replace X_t in the yield and return regressions with the fitted value $\hat{X}_t$.

Interpretation: The IV strategy addresses the concern that Treasury maturity policy could respond to investor demand. Debt-to-GDP is correlated with maturity-weighted debt but is plausibly driven mostly by accumulated deficits rather than contemporaneous demand for long bonds.

4.6 Arbitrageur-wealth interaction

The paper constructs two measures of lagged arbitrageur wealth changes:

$$\Delta W_{1t}^{Arb} = \left(y_{t-1}^{(\tau)} - y_{t-1}^{(1)} \right) r x_t^{(\tau)},$$

$$\Delta W_{2t}^{Arb} = \left(\frac{MWD}{GDP} \right)_{t-1} r x_t^{(\tau)}.$$

The return regression is augmented with interactions between supply and these wealth changes:

$$r_{t+k,k}^{(LT)} = a + bX_t + e\Delta W_t^{Arb}X_t + \dots + u_{t+k}.$$

Interpretation: Negative interaction coefficients mean that supply effects are stronger after arbitrageurs lose money. This supports the slow-moving-capital interpretation: when arbitrageurs have less wealth or higher effective risk aversion, supply shocks require larger expected-return compensation.

5 Identification and empirical design

5.1 What the paper is trying to identify

The paper tries to identify whether changes in the quantity and maturity structure of Treasury debt cause changes in bond yields and expected returns. The central empirical challenge is that Treasury supply is not randomly assigned. Governments may issue different maturities in response to interest-rate conditions, budget needs, or investor demand.

Interpretation: The paper therefore does not rely on one design alone. It triangulates across model predictions, time-series regressions, historical samples, IV tests, robustness controls, and model calibration.

5.2 Predetermined supply and face-value measurement

The paper uses face-value cash flows rather than market values:

$$D_t^{(\tau)} = \text{promised cash flow at maturity } \tau.$$

- Market values would move mechanically with yields and returns.
- A rise in yields lowers market values, mechanically lowering measured supply if supply were market-value based.
- Face-value measurement reduces this mechanical negative relationship.

Interpretation: This measurement choice is essential. If supply were measured with market values, the dependent and independent variables would be mechanically connected through prices.

5.3 Short-rate control

All baseline yield and return regressions control for $y_t^{(1)}$:

$$y_t^{(\tau)} = a + bX_t + cy_t^{(1)} + u_t.$$

- The short rate captures monetary-policy conditions and the level of the yield curve.
- Supply is negatively correlated with the short rate in the data.
- Omitting the short rate could confound supply effects with monetary-policy effects.

Interpretation: The paper is not asking whether high-debt periods have high or low interest rates in general. It asks whether, conditional on the short rate, greater duration supply raises long yields and expected returns.

5.4 Instrumental variables design

The IV design uses marketable Treasury debt-to-GDP as an instrument for MWD/GDP .

The logic is:

$$\frac{D}{GDP} \rightarrow \frac{MWD}{GDP} \rightarrow \text{yields and returns.}$$

- The first stage is strong: debt-to-GDP explains a large share of the variation in maturity-weighted debt.
- The instrument is motivated by the idea that debt-to-GDP is driven mainly by accumulated fiscal deficits.
- The concern it addresses is reverse causality or maturity timing: governments may choose maturity to exploit investor demand conditions.

Interpretation: The exclusion restriction is that debt-to-GDP affects bond risk premia mainly through maturity-weighted supply, conditional on the short rate and controls. This assumption is plausible but not mechanically guaranteed, so the paper treats the IV as one piece of a broader robustness strategy.

5.5 Historical validation using the prewar sample

The paper repeats the main analysis in the 1916–1940 sample:

$$\begin{aligned} y_t^{(LT)} &= a + b(MWD/GDP)_t + cy_t^{(1)} + u_t, \\ r_{t+k,k}^{(LT)} &= a + b(MWD/GDP)_t + cy_t^{(1)} + u_{t+k}. \end{aligned}$$

- The sample uses annual data rather than monthly data.
- It predates the postwar institutional environment.
- The relation between supply and three-year long-bond returns remains strongly positive.

Interpretation: Replication in an earlier period reduces concerns that the main postwar result is driven by one institutional regime or a few postwar monetary episodes.

5.6 Maturity-gradient validation

The maturity-gradient prediction is a powerful internal validity check. If the coefficient of supply is truly a duration-risk effect, it should be larger for longer maturities.

The paper therefore estimates yield and return regressions for multiple maturities:

$$\{2, 3, 4, \dots, 15\}\text{-year bonds.}$$

Interpretation: A generic omitted variable might affect all bond maturities similarly. The model predicts a specific maturity pattern. Finding that supply effects rise with maturity is therefore more diagnostic than simply finding a positive average coefficient.

5.7 Arbitrageur capital validation

The model predicts that supply effects should be stronger when arbitrageurs are more risk averse or have recently lost wealth. The paper tests this by interacting supply with proxies for lagged arbitrageur wealth changes.

Interpretation: This is a second internal validation check. If supply effects arise because arbitrageurs must absorb duration risk, the effect should be state dependent: the same supply shock should matter more when arbitrageurs' risk-bearing capacity is low.

5.8 Threats to identification

- **Endogenous maturity choice.** The Treasury may issue more long-term debt when investor demand for long duration is high. This would bias supply effects downward and motivates IV tests.
- **Macro confounds.** Inflation, growth, term spreads, and risk volatility may affect both debt policy and expected returns. The paper adds each as a control.
- **Mechanical market-value effects.** Market-value supply measures mechanically move with yields. The main analysis uses face-value cash flows.
- **Regime dependence.** Monetary regimes and Fed policies change over time. The paper uses both postwar and prewar samples and omits the Fed-peg years.
- **Small sample and overlapping returns.** Long-horizon returns overlap and create serial correlation. The paper reports Newey-West and AR/ARMA-based inference.

Interpretation: The empirical design is best read as a cumulative case. The IV test is important, but the strongest evidence comes from the alignment of many patterns with the model: positive effects, maturity gradients, robustness, historical replication, and arbitrageur-wealth interactions.

6 Tables and figures

6.1 Figure 1: Principal and coupon payments; Figure 2: Bond supply, 1952–2007

Read:

- Figure 1 shows that much of the Treasury cash-flow profile is concentrated at short maturities, but a long right tail of coupon and principal payments extends across the maturity spectrum.
- Figure 2 shows that MWD/GDP , LTD/GDP , average maturity M , and debt-to-GDP move substantially over time.
- The major rise in maturity-weighted supply from the mid-1970s to early 1990s is central to the empirical variation.

Interpretation: Figure 1 motivates why a duration-weighted measure is needed: two debt structures with the same face value can impose different duration risk on arbitrageurs. Figure 2 shows that there is enough low-frequency variation in Treasury maturity structure to test the model.

Economic message: The pair connects measurement to economic content: debt supply is not just a level, but a schedule of future payments whose timing determines duration risk.

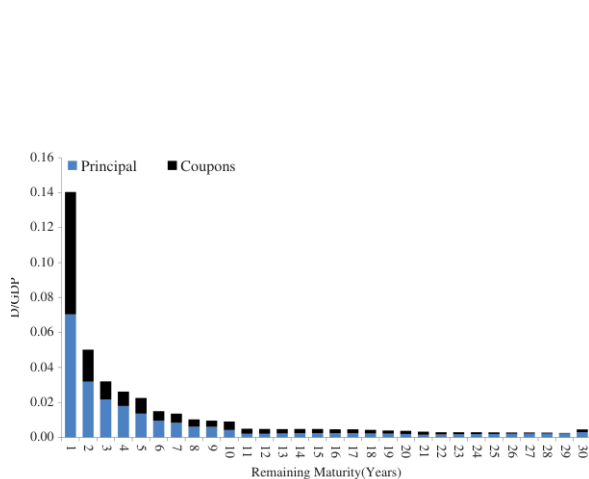


Figure 1: Principal and coupon payments
The time-series average, for each maturity year, of total payments on bills, bonds, and notes scaled by GDP. The bottom bars denote principal payments. The darker top bars denote coupon payments. Any payments beyond 30 years are included in the 30-year bucket. The data are based on the CRSP bond database and cover the period 1952–December 2007.

(a) **Figure 1: Principal and coupon payments.** The maturity distribution of Treasury cash flows shows the amount of principal and coupon payments due at each future maturity, scaled by GDP.

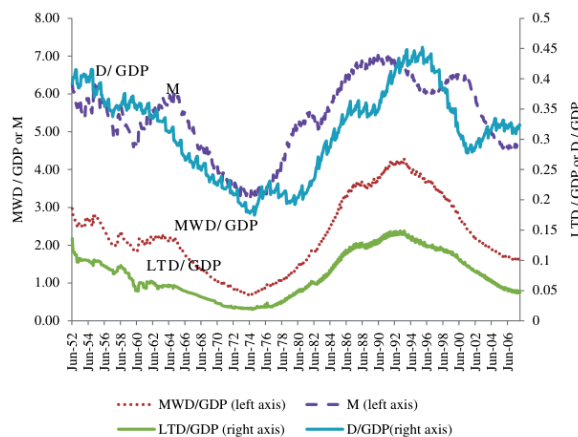


Figure 2: Bond supply, 1952–2007
 MWD/GDP is the maturity-weighted-debt-to-GDP ratio, computed by multiplying each debt payment by the corresponding maturity, summing across maturities, and scaling by GDP. LTD/GDP is the long-term-debt-to-GDP ratio, computed by summing all debt payments with maturity beyond ten years, and scaling by GDP. M is the dollar-weighted average maturity expressed in years. D/GDP is the ratio of the aggregate principal payments of all Treasury securities to GDP. MWD/GDP , LTD/GDP , and M are computed using aggregate principal and coupon payments.

(b) **Figure 2: Bond supply, 1952–2007.** Time-series variation in maturity-weighted debt, long-term debt, average maturity, and debt-to-GDP.

6.2 Table 1: Summary Statistics; Figure 3: Bond supply, 1916–1940

Read:

- Table 1 reports that MWD/GDP has mean 2.284 and standard deviation 0.997 in the 1952–2007 sample.
- The long-term yield has mean 6.3%, and long-bond one-, three-, and five-year returns show large dispersion.
- Panel B shows that the prewar sample has higher average maturity M and a higher mean MWD/GDP .
- Figure 3 shows large swings in prewar maturity-weighted supply, especially around World War I and the interwar period.

Interpretation: Table 1 calibrates the size of the variables: a one-standard-deviation MWD/GDP shock is economically meaningful. Figure 3 shows that the prewar sample offers an independent source of variation that is not simply a replica of the postwar Treasury market.

Economic message: The table and figure together show that the paper has both cross-time variation and economically large variation in the key state variable.

Table 1
Summary Statistics

	Mean	Median	SD	Min	Max
Panel A. Main sample 1952–2007					
Measures of Debt Maturity and Debt Supply:					
MWD/GDP	2.284	2.142	0.997	0.671	4.275
MWD/GDP (MV-based)	1.050	1.117	0.444	0.338	2.127
LTD/GDP	0.077	0.070	0.038	0.019	0.149
LTD/GDP (MV-based)	0.047	0.046	0.025	0.011	0.116
M	5.387	5.511	1.016	3.269	7.024
D/GDP	0.313	0.322	0.073	0.175	0.452
Returns and Yields:					
$y^{(LT)}$	0.063	0.060	0.025	0.026	0.138
$y^{(LT)} - y^{(1)}$	0.008	0.008	0.012	-0.032	0.041
$y^{(1)}$	0.055	0.053	0.029	0.006	0.158
r_1	0.062	0.049	0.093	-0.187	0.434
r_3	0.189	0.187	0.165	-0.187	0.680
r_5	0.320	0.312	0.232	-0.166	1.100
Macroeconomic Conditions and Other Controls:					
Output gap	0.000	0.000	0.016	-0.048	0.038
Output growth	0.033	0.033	0.024	-0.031	0.091
Inflation	0.038	0.031	0.028	-0.009	0.138
Inflation risk	0.002	0.002	0.001	0.001	0.006
Interest-rate risk	0.007	0.006	0.006	0.001	0.035
Stock-market risk	0.007	0.006	0.004	0.002	0.049
MWD/GDP (Fed adjusted)	2.154	2.044	1.001	0.555	4.134
Arbitrageur Wealth:					
ΔW^{Arb}	0.000	0.000	0.001	-0.004	0.007
ΔW_2^{Arb}	0.040	0.006	0.225	-0.673	0.958
Panel B. Prewar sample 1916–1940					
MWD/GDP	3.062	3.500	0.974	0.552	4.264
LTD/GDP	0.128	0.151	0.045	0.022	0.181
M	12.857	12.307	3.572	9.153	23.377
D/GDP	0.250	0.238	0.104	0.024	0.426
$y^{(LT)}$	0.035	0.036	0.008	0.022	0.055
$y^{(1)}$	0.023	0.029	0.019	0.000	0.058
r_1	0.000	0.000	0.000	0.000	0.001
r_3	0.031	0.035	0.010	0.006	0.043

Panel A summarizes the main sample 1952–2007. MWD/GDP is the maturity-weighted-debt-to-GDP ratio. Its market-value-based version multiplies the market value of each bond by Macaulay duration, sums across bonds, and scales by GDP. LTD/GDP is the long-term-debt-to-GDP ratio. Its market-value-based version sums the market values of all bonds with maturity beyond ten years, and scales by GDP. M is the dollar-weighted average maturity expressed in years. D/GDP is the ratio of the aggregate principal payments of all Treasury securities to GDP. $y^{(LT)}$ is the yield of a long-term bond with approximate maturity twenty years, and $y^{(1)}$ is the one-year yield. r_1 , r_2 , and r_3 are holding-period returns for the long-term bond over one-, three-, and five-year horizons, respectively. Output gap is the residual from a Hodrick-Prescott filter of log GDP. Output growth is the difference between log real GDP in the most recent quarter t and log real GDP in quarter $t-4$. Inflation risk is the standard deviation of monthly inflation over the most year. Interest-rate risk is the standard deviation of the

(a) **Table 1: Summary Statistics.** Distribution of supply measures, yields, returns, macro controls, and arbitrageur-wealth proxies in the main and prewar samples.

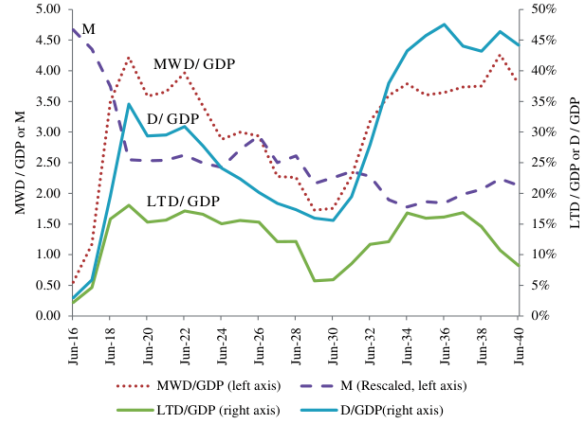


Figure 3
Bond supply, 1916–1940
 MWD/GDP is the maturity-weighted-debt-to-GDP ratio, computed by multiplying each debt payment by the corresponding maturity, summing across maturities, and scaling by GDP. LTD/GDP is the long-term-debt-to-GDP ratio, computed by summing all debt payments with maturity beyond ten years, and scaling by GDP. M is the dollar-weighted average maturity expressed in years and divided by five (to fit into the picture). D/GDP is the ratio of the aggregate principal payments of all Treasury securities to GDP. MWD/GDP , LTD/GDP , and M are computed using aggregate principal payments.

(b) **Figure 3: Bond supply, 1916–1940.** Prewar analog of the supply measures used for robustness.

6.3 Table 2: Bond supply, bond yields, and bond returns; Figure 4: Bond supply and excess bond returns

Read:

- Table 2 is the main evidence: MWD/GDP positively predicts yields and future returns after controlling for the one-year yield.
- The long-term yield coefficient on MWD/GDP is 0.004, and the one-, three-, and five-year long-bond return coefficients are 0.026, 0.065, and 0.094.
- Results are similar using LTD/GDP , though magnitudes differ because the variable is scaled differently.
- Figure 4 visualizes the key return relation: high maturity-weighted supply is associated with high subsequent three-year long-bond excess returns in both samples.

Interpretation: Table 2 establishes the basic portfolio-balance result. The yield results show contemporaneous price impact; the return results show predictability of risk premia. Figure 4 makes clear that the relationship is not only a regression artifact: the positive slope appears in the scatterplots for both the main and prewar samples.

How it fits the paper: The model predicts positive yield and expected-return coefficients. Table 2 and Figure 4 provide the first and most direct empirical support for those predictions.

Table 2
Bond supply, bond yields, and bond returns

	X = MWD/GDP							X = LTD/GDP						
	b	[t NW]	[t AR]	c	[t NW]	[t AR]	R ²	b	[t NW]	[t AR]	c	[t NW]	[t AR]	R ²
Yield spreads:														
Yield 2-yr bond	0.001	[2.597]	[2.363]	0.981	[50.113]	[70.970]	0.987	0.029	[2.476]	[2.293]	0.982	[49.381]	[69.955]	0.987
Yield 3-yr bond	0.002	[2.510]	[1.881]	0.951	[29.510]	[36.999]	0.968	0.044	[2.364]	[1.811]	0.952	[29.150]	[36.591]	0.968
Yield 4-yr bond	0.002	[2.497]	[1.805]	0.932	[22.657]	[30.645]	0.949	0.058	[2.356]	[1.772]	0.934	[22.419]	[30.362]	0.948
Yield 5-yr bond	0.002	[2.358]	[1.580]	0.913	[19.528]	[23.621]	0.933	0.064	[2.258]	[1.601]	0.914	[19.387]	[23.506]	0.932
Yield LT bond	0.004	[2.682]	[1.719]	0.795	[12.167]	[12.993]	0.379	0.107	[2.610]	[1.822]	0.797	[12.234]	[13.253]	0.374
Returns:														
1-year return 2-yr bond	0.004	[1.979]	[1.438]	1.114	[12.201]	[11.290]	0.774	0.116	[2.176]	[1.590]	1.118	[12.214]	[11.245]	0.776
1-year return 3-yr bond	0.007	[1.860]	[1.512]	1.134	[6.751]	[6.847]	0.507	0.191	[2.013]	[1.627]	1.140	[6.750]	[6.804]	0.509
1-year return 4-yr bond	0.010	[1.964]	[1.774]	1.157	[4.864]	[4.535]	0.358	0.266	[2.084]	[1.867]	1.166	[4.855]	[4.495]	0.360
1-year return 5-yr bond	0.011	[1.902]	[1.852]	1.145	[3.897]	[4.172]	0.263	0.308	[2.012]	[1.913]	1.154	[3.897]	[4.132]	0.265
1-year return LT bond	0.026	[3.097]	[3.462]	1.212	[2.846]	[3.181]	0.190	0.685	[3.196]	[3.468]	1.229	[2.860]	[3.142]	0.189
3-year return LT bond	0.065	[4.200]	[4.121]	3.737	[4.971]	[4.587]	0.506	1.786	[4.200]	[4.284]	3.795	[5.039]	[4.627]	0.516
5-year return LT bond	0.094	[5.421]	[3.580]	6.139	[5.401]	[4.650]	0.658	2.625	[5.340]	[4.068]	6.235	[5.612]	[5.062]	0.675

Monthly time-series regressions of the form:

$$y_t^{(\tau)} = a + bX_t + cy_t^{(1)} + u_t$$

$$r_{t+k,k}^{(\tau)} = a + bX_t + cy_t^{(1)} + u_{t+k}$$

The dependent variable is the yield or the one-year, three-year, or five-year return of the τ -year bond. The independent variable X_t is MWD/GDP , the maturity-weighted-debt-to-GDP ratio, or LTD/GDP , the long-term-debt-to-GDP ratio. The regressions control for the one-year yield. The first set of t -statistics, reported in brackets, is based on Newey-West standard errors, with 36 lags in the case of the yield and one-year return regressions, and 54 and 90 lags in the case of the three- and five-year return regressions. The second set of t -statistics is based on modeling the error process as AR(1) for the yield regressions, and as ARMA(1, k) for the return regressions where k denotes the number of months in the return cumulation (e.g., twelve for the one-year return).

Table 2: Bond supply, bond yields, and bond returns. Monthly regressions of yields and future bond returns on MWD/GDP or LTD/GDP , controlling for the one-year yield.

6.4 Table 3: Regression coefficients as a function of maturity; Figure 5: Regression coefficients as a function of maturity

Read:

- Table 3 estimates the supply coefficient separately for zero-coupon maturities from two to fifteen years.
- Yield coefficients rise slowly with maturity, from 0.0009 at two years to 0.0032 at fifteen years.
- Return coefficients rise more strongly, from 0.005 at two years to 0.039 at fifteen years.
- Figure 5 plots the same maturity pattern with confidence intervals.

Interpretation: This pair is a key model-validation test. If supply works through duration risk, longer maturities should react more. The return pattern is especially consistent with the model because long bonds are most exposed to changes in the price of duration risk.

Economic message: The maturity-gradient evidence rules out a simple level-of-rates explanation. The model predicts not only that supply matters, but also how its effect should vary across maturities.

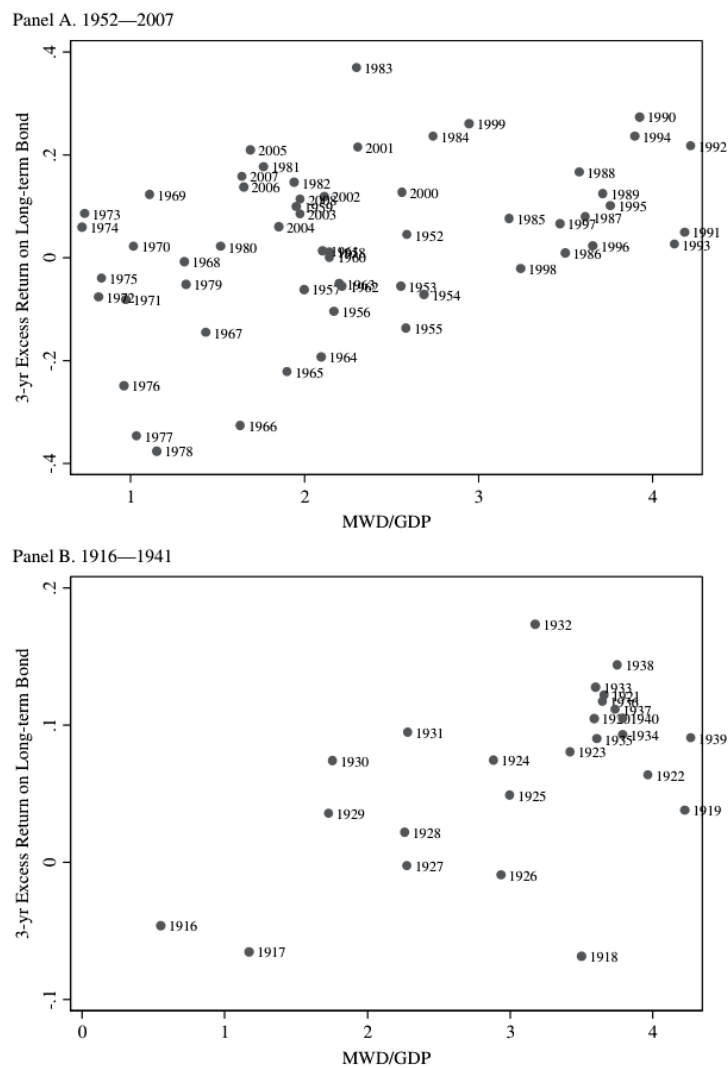


Figure 4
Bond supply and excess bond returns
 Plots of three-year holding-period excess return on long-term government bonds (vertical axis) against the maturity-weighted-debt-to-GDP ratio (horizontal axis). Panel A shows the 1952–2007 period. Panel B shows the 1916–1941 period.

Figure 4: Bond supply and excess bond returns. Scatterplots of three-year long-bond excess returns against MWD/GDP for 1952–2007 and 1916–1941.

Table 3
Bond supply, bond yields, and bond returns: Regression coefficients as a function of maturity

	Yields				1-yr returns			
	b	[t NW]	[t AR]	R^2	b	[t NW]	[t AR]	R^2
$\tau =$								
2-yr	0.0009	[2.051]	[1.567]	0.985	0.005	[2.143]	[1.484]	0.752
3-yr	0.0014	[2.016]	[1.360]	0.961	0.008	[2.134]	[1.659]	0.489
4-yr	0.0017	[2.045]	[1.303]	0.938	0.011	[2.223]	[1.881]	0.339
5-yr	0.0020	[2.087]	[1.280]	0.916	0.014	[2.340]	[2.088]	0.254
6-yr	0.0023	[2.133]	[1.271]	0.898	0.016	[2.459]	[2.267]	0.203
7-yr	0.0025	[2.178]	[1.270]	0.882	0.019	[2.569]	[2.421]	0.171
8-yr	0.0026	[2.222]	[1.275]	0.868	0.022	[2.663]	[2.555]	0.151
9-yr	0.0028	[2.263]	[1.286]	0.856	0.024	[2.740]	[2.671]	0.137
10-yr	0.0029	[2.301]	[1.301]	0.846	0.027	[2.800]	[2.772]	0.128
11-yr	0.0030	[2.334]	[1.319]	0.836	0.029	[2.845]	[2.858]	0.121
12-yr	0.0030	[2.363]	[1.338]	0.827	0.032	[2.878]	[2.933]	0.115
13-yr	0.0031	[2.386]	[1.356]	0.819	0.034	[2.901]	[2.998]	0.111
14-yr	0.0032	[2.403]	[1.373]	0.812	0.037	[2.916]	[3.054]	0.107
15-yr	0.0032	[2.414]	[1.388]	0.805	0.039	[2.926]	[3.104]	0.104

Monthly time-series regressions of the form:

$$y_t^{(\tau)} = a + b_\tau(MWD/GDP)_t + c_\tau^{(1)} + u_t, \text{ and}$$

$$r_{t+k,k}^{(\tau)} = a + b_\tau(MWD/GDP)_t + c_\tau^{(1)} + u_{t+k}.$$

for each maturity τ between two and fifteen years, during the 1971–2007 period. The dependent variable is the yield or the one-year return of the τ -year bond. The independent variable is MWD/GDP , the maturity-weighted-debt-to-GDP ratio. The regressions control for the one-year yield. The first set of t -statistics, reported in brackets, is based on Newey-West standard errors with 36 lags. The second set of t -statistics is based on modeling the error process as AR(1) for the yield regressions, and as ARMA(1,12) for the return regressions.

(a) **Table 3: Regression coefficients as a function of maturity.** Yield and one-year-return regressions by maturity, 1971–2007.

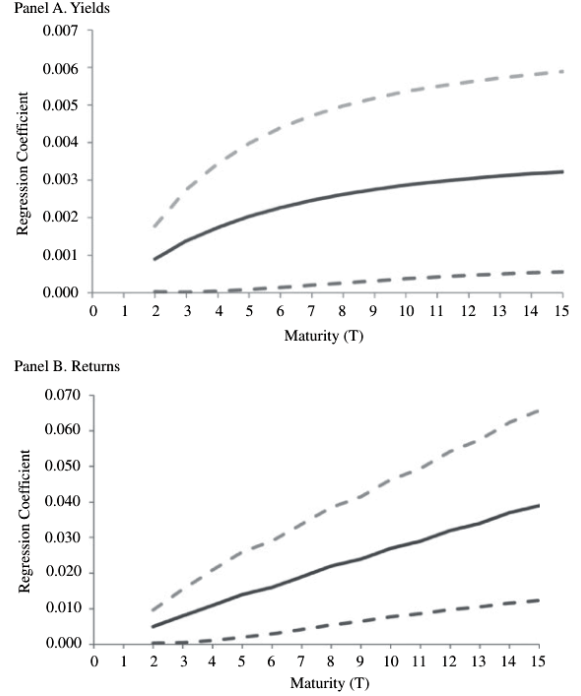


Figure 5
Bond supply, bond yields, and bond returns: Regression coefficients as a function of maturity
We use the data provided by Gürkaynak, Sack, and Wright (2007) to obtain yields and one-year returns for zero-coupon bonds with maturities between two and fifteen years during the November 1971–December 2007 period. For each maturity τ , we estimate monthly time-series regressions of the form:

$$y_t^{(\tau)} = a + b_\tau(MWD/GDP)_t + c_\tau^{(1)} + u_t, \text{ and}$$

$$r_{t+k,k}^{(\tau)} = a + b_\tau(MWD/GDP)_t + c_\tau^{(1)} + u_{t+k}.$$

(b) **Figure 5: Regression coefficients as a function of maturity.** Visual maturity profile of supply coefficients and confidence bands.

6.5 Table 4: Instrumental-variables tests; Table 5: Robustness tests

Read:

- Table 4 instruments MWD/GDP with debt-to-GDP. The first-stage R^2 is 0.835.
- IV return coefficients are 0.028 for one-year returns and 0.067 for three-year returns, very close to the corresponding OLS coefficients.
- Table 5 shows robustness to market-value supply measures, macro controls, term-spread controls, future debt maturity, Fed holdings, and alternative sampling months.
- The supply coefficient remains positive and generally statistically significant across these specifications.

Interpretation: Table 4 addresses endogenous maturity choice. If the Treasury issues long debt when long bonds are expensive, OLS could be biased downward. The IV results show that the return relation survives. Table 5 shows that the result is not absorbed by standard macro predictors or by alternative measurement choices.

How it fits the paper: These tables are the paper’s main robustness block. They show that the positive supply-return relation is not an artifact of market-value measurement, business-cycle variables, or the timing

of observations.

Table 4

Instrumental-variables tests

First-stage regressions: $X_t = (MWD/GDP)_t = \alpha + \beta(D/GDP)_t + \delta y_t^{(1)} + u_t$

Instrument = D/GDP			
D/GDP	14.186		
	[12.33]		
$y^{(1)}$	12.809		
	[4.08]		
R^2	0.835		
Second-stage regressions: $y_t^{(LT)} = a + bX_t + y_t^{(1)} + u_t$, and			
	$r_{t+k}^{(LT)} = a + bX_t + y_t^{(1)} + u_{t+k}$		
	Yield	1-year return	3-year return
MWD/GDP (IV)	0.002	0.028	0.067
	[1.129]	[2.885]	[3.493]
$y^{(1)}$	0.845	1.223	3.748
	[11.600]	[2.868]	[4.923]
R^2	0.849	0.189	0.506

We instrument for the maturity-weighted-debt-to-GDP ratio by the Treasury-debt-to-GDP ratio. Both the first- and second-stage regressions are monthly and include a control for the one-year yield. The dependent variable in the second-stage regression is the yield or the one- or three-year return of the long-term bond. t -statistics, reported in parentheses, are based on Newey-West standard errors with 36 lags in the case of the yield and 60 lags in the case of the returns.

(a) **Table 4: Instrumental-variables tests.** Debt-to-GDP instruments for maturity-weighted debt-to-GDP in yield and return regressions.

Table 5
Robustness tests

	1-year return			3-year return		
	b	[t]	R^2	b	[t]	R^2
(1) Base case: control for short rate only	0.026	[3.097]	0.190	0.065	[4.200]	0.506
Market-value-based measures of supply:						
(2) MWD/GDP MV (OLS)	0.003	[1.748]	0.136	0.011	[2.269]	0.426
(3) MWD/GDP MV (IV)	0.007	[2.671]	0.114	0.016	[3.110]	0.409
(4) LTD/GDP MV (OLS)	0.894	[2.586]	0.155	2.887	[3.728]	0.489
(5) LTD/GDP MV (IV)	1.395	[2.837]	0.142	3.317	[3.448]	0.486
Macroeconomic and other controls:						
(6) Add control for term spread	0.014	[1.892]	0.281	0.046	[2.422]	0.582
(7) Add control for output gap	0.026	[3.399]	0.207	0.065	[4.308]	0.536
(8) Add control for output growth	0.026	[3.165]	0.188	0.066	[4.376]	0.522
(9) Add control for inflation	0.020	[2.680]	0.210	0.061	[4.511]	0.509
(10) Add control for inflation risk	0.027	[3.124]	0.191	0.076	[4.809]	0.588
(11) Add control for interest-rate risk	0.026	[3.126]	0.189	0.068	[4.358]	0.519
(12) Add control for stock-market risk	0.026	[3.101]	0.194	0.064	[4.324]	0.538
(13) Time trend control	0.020	[2.343]	0.215	0.045	[2.892]	0.594
(14) Future changes in debt maturity	0.027	[3.198]	0.202	0.067	[4.402]	0.524
(15) Adjust MWD for Fed holdings	0.027	[3.120]	0.199	0.660	[4.242]	0.516
(16) Annual sampling (September)	0.028	[3.085]	0.183	0.072	[4.106]	0.505
(17) Annual sampling (December)	0.023	[2.524]	0.139	0.060	[3.538]	0.456

OLS and instrumental-variables monthly time-series regressions of the form:

$$r_{t+k}^{(LT)} = a + b(MWD/GDP)_t + c y_t^{(1)} + d Z_t + u_{t+k}$$

where Z denotes a control that includes the term spread, output gap, output growth, inflation, inflation risk, interest-rate risk and stock-market risk. The dependent variable is the one- or three-year return of the long-term bond. t -statistics, reported in parentheses, are based on Newey-West standard errors, with 36 lags in the case of the yield and 60 lags in the case of the returns.

(b) **Table 5: Robustness tests.** Alternative supply measures, macro controls, Fed holdings, and annual sampling.

6.6 Table 6: Bond supply, yields, and returns, 1916–1940; Table 7: Arbitrageur wealth

Read:

- Table 6 repeats the main long-bond regressions in the prewar sample. The three-year return coefficient is 0.029 with a large t -statistic.
- Table 7 tests whether supply effects are stronger when arbitrageurs have recently lost money.
- The interaction terms are negative in the relevant specifications, meaning that a decline in arbitrageur wealth amplifies the effect of supply on returns.
- The direct MWD/GDP coefficient remains positive in the one- and three-year return regressions.

Interpretation: Table 6 is a historical external-validity test. Table 7 is a mechanism test. The supply-return relation is stronger when the agents who absorb duration supply are plausibly more constrained or more risk averse.

Economic message: The prewar and arbitrageur-wealth results jointly strengthen the portfolio-balance interpretation: the effect is not period-specific, and it varies with risk-bearing capacity.

6.7 Figure 6: Model calibration

Read:

- Figure 6 compares empirical coefficients from Table 3 with model-implied coefficients.
- The model reproduces the broad ordering: return effects are larger than yield effects, and both rise with maturity.
- The empirical return curve rises faster than the model curve at long maturities, but the calibrated model captures the main qualitative pattern.

Table 6
Bond supply, bond yields, and bond returns, 1916–1940

	Yield	Returns	
		1-year return	3-year return
MWD/GDP	0.001 [1.310]	0.008 [0.988]	0.029 [5.528]
$y^{(1)}$	0.379 [5.216]	0.035 [0.103]	1.169 [2.778]
R^2	0.698	0.043	0.448

Annual time-series regressions of the form:

$$y_t^{(LT)} = a + bMWD/GDP_t + c y_t^{(1)} + u_t$$

$$r_{t+k}^{(LT)} = a + bMWD/GDP_t + c y_t^{(1)} + u_{t+k}$$

during the 1916–1940 period (with one-year returns extending to 1941 and three-year returns extending to 1943). The data are sampled at the end of June. The dependent variable is the yield or the one- or three-year return of the long-term bond. t -statistics, reported in parentheses, are based on Newey-West standard errors, with three lags in the case of the yield and one-year return regressions, and five lags in the case of the three-year return regression.

Table 6: Bond supply, bond yields, and bond returns, 1916–1940. Annual prewar regressions for the long-term bond.

Interpretation: Figure 6 translates the reduced-form coefficients back into the model. It shows that a term-structure model with risk-averse arbitrageurs can generate supply effects of the right sign and rough magnitude.

How it fits the paper: This final figure closes the loop between theory and evidence: the same mechanism used to derive the regressions can be calibrated to match their cross-maturity shape.

7 Short summary

- The paper studies whether the maturity structure of U.S. government debt affects bond yields and expected returns.
- It builds a term-structure model in which risk-averse arbitrageurs absorb duration-supply shocks.
- The main empirical measure is maturity-weighted debt to GDP, MWD/GDP .
- Higher MWD/GDP predicts higher bond yields and higher future bond returns.
- Effects are stronger for longer maturities, especially for future returns.
- IV regressions using debt-to-GDP as an instrument produce similar return coefficients.
- Robustness tests show that the supply-return relation survives macro controls, term-spread controls, Fed holdings adjustments, and alternative supply measures.
- A prewar 1916–1940 sample produces similar results for long-bond returns.
- Supply effects are stronger when arbitrageurs have recently lost money.
- Model calibration broadly matches the observed maturity pattern of yield and return coefficients.

8 Conclusion

8.1 Core conclusion

Greenwood and Vayanos show that the supply and maturity structure of government debt affect the term structure of interest rates. In their evidence, a larger supply of duration-weighted Treasury debt raises long-term yields and forecasts higher future bond returns, even after controlling for the short rate. This contradicts strict maturity irrelevance and supports portfolio-balance theories.

Table 7
Bond supply, bond returns, and arbitrageur wealth

	1-year return				3-year return			
	0.026 [3.102]	0.031 [3.408]	0.034 [3.634]	3.443 [5.136]	4.26 [5.190]	4.114 [5.633]	0.063 [4.210]	0.067 [4.252]
MWD/GDP								
$y^{(LT)} - y^{(1)}$								
$\Delta W_t^{LAB} MWD/GDP$		-4.729 [-3.380]					-1.971 [-1.221]	
$\Delta W_t^{LAB} MWD/GDP$			-0.029 [-3.114]					-0.013 [-1.422]
$\Delta W_t^{LAB} y^{(LT)} - y^{(1)}$				-675.811 [-2.774]				-509.326 [-2.404]
$\Delta W_t^{LAB} y^{(LT)} - y^{(1)}$					-5.226 [-2.954]			-2.792 [-1.153]
$y^{(1)}$	1.214 [2.791]	1.146 [2.587]	1.130 [2.489]	1.894 [5.405]	1.838 [5.408]	1.909 [5.918]	3.741 [4.830]	3.713 [4.806]
R^2	0.190	0.239	0.238	0.265	0.309	0.307	0.503	0.505

Monthly time-series regressions of the form:

$$r_{t+k}^{(LT)} = a + bMWD/GDP_t + c y_t^{(LT)} - y_t^{(1)} + e \Delta W_t^{LAB} MWD/GDP_t + f \Delta W_t^{LAB} y_t^{(LT)} - y_t^{(1)} + e y_t^{(1)} + u_{t+k}$$

The dependent variable is the one- or three-year return of the long-term bond. The independent variables include the spread between the long-term and the one-year yield, MWD/GDP, and interactions between these variables and changes in arbitrageur wealth. We use two measures of arbitrageur wealth. The first measure, ΔW_t^{LAB} , is the product of the one-year lagged spread between the long-term and the one-year yield, times the subsequent one-year excess return of the long-term bond. The second measure, ΔW_t^{LAB} , is the product of the one-year lagged MWD/GDP times the subsequent one-year excess return of the long-term bond. t -statistics, reported in parentheses, are based on Newey-West standard errors, with 36 lags in the case of the

Table 7: Bond supply, bond returns, and arbitrageur wealth. Interactions between supply and lagged arbitrageur-wealth measures.

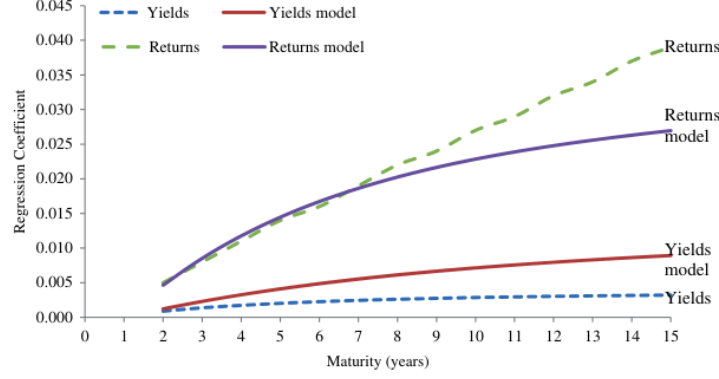


Figure 6
Model calibration

The solid and dashed lines plot the coefficients b_τ from the monthly time-series regressions:

$$y_t^{(\tau)} = a + b_\tau (MWD/GDP)_t + c y_t^{(1)} + u_t, \text{ and}$$

$$r_{t+k,k}^{(\tau)} = a + b_\tau (MWD/GDP)_t + c y_t^{(1)} + u_{t+k},$$

as a function of maturity τ ranging from two to fifteen years. In the first equation the yield of a zero-coupon bond with maturity τ is regressed on the maturity-weighted-debt-to-GDP ratio, controlling for the one-year yield. In the second equation the dependent variable is the bond's one-year return. The solid lines are the coefficients derived from the model. The dashed lines are the coefficients when the regressions are performed on actual data during the 1971–2007 period. The risk aversion coefficient of arbitrageurs in the model is chosen so that the average across all points in the two solid lines is equal to the average across all points in the two dashed lines.

Figure 6: Model calibration. Empirical and model-implied supply coefficients for yields and one-year returns across maturities.

8.2 Economic intuition

The intuition is that long-term bonds expose investors to duration risk. When the government issues more long-duration debt, private arbitrageurs must hold more interest-rate risk. Because they are risk averse, they require compensation. That compensation appears as lower bond prices, higher yields, and higher expected excess returns. The effect is strongest for long-maturity bonds because they are most sensitive to changes in the price of duration risk.

8.3 Identification takeaway

The paper's identification is not based on a single quasi-experiment. It is based on a model-guided empirical pattern. The main evidence is a positive relation between MWD/GDP and future returns. The stronger evidence is that the relation has the exact patterns predicted by the model: it is stronger for longer maturities, survives an IV strategy and macro controls, appears in a prewar sample, and becomes stronger when arbitrageurs have recently lost money.

8.4 Policy implication

The results imply that debt-management policy and central-bank asset-purchase programs can influence long-term rates by changing the duration risk borne by the private sector. If the government or central bank removes duration from the market, long yields and expected returns should fall. If it issues more long-duration debt, long yields and risk premia should rise.

8.5 Limitations

The paper's time-series design cannot fully eliminate all endogeneity concerns. Debt-to-GDP may affect macroeconomic conditions through channels other than maturity-weighted supply, and Treasury maturity

decisions may still respond to investor demand. The paper addresses these issues through IV, controls, and multiple samples, but the estimates should be read as model-guided evidence rather than a randomized experiment.

8.6 Takeaway

The paper's lasting contribution is to make the portfolio-balance channel empirically disciplined. It shows that duration-weighted bond supply has predictive power for bond yields and expected returns, and it connects those predictive patterns to a structural mechanism in which risk-averse arbitrageurs price duration risk.